Assignment Day 1
C++ Programming

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1. Print Hello World:

Program :

```
#include<iostream>
using namespace std;

int main(){
    cout << "Hello World";
    return 0;
}
```

Output:

```
(base) abhinyaay@Abhinays-MacBook-Air Desktop % cd cpp\
(base) abhinyaay@Abhinays-MacBook-Air cpp  % ./hello.0
Hello World%
(base) abhinyaay@Abhinays-MacBook-Air cpp  %
```

2. Add two numbers:

Program:

```
#include<iostream>
using namespace std;

int main (){
    int num1, num2,result;

    cout << "Enter number 1 :" ;
    cin >> num1;
    cout << "Enter number 2 :" ;
```

```
    cin >> num2;

    result = num1 + num2;

    cout << "Addtion of number " << num1 << " and " <<num2 <<"
is " <<result<<endl;

    return 0;

}
```

Output:

```
(base) abhinyaay@Abhinays-MacBook-Air cpp  % ./addnumber.o
Enter number 1 :20
Enter number 2 :45
Addtion of number 20 and 45 is 65
(base) abhinyaay@Abhinays-MacBook-Air cpp  %
```

Add two binary numbers:

Program:

```
#include<iostream>
using namespace std;

int powercal(int base, int expo){
    int num = 1;
    for( int i=0; i<expo ; i++ ){
        num = num * base;
    }
    return num;
}
// funtion to convert binary to decimal
int binaryToDecimal(int binary_num){
    int decimal, digit, power;
```

```
    decimal = 0;
    power = 0;
    while (binary_num > 0){
        digit = binary_num % 10;
        decimal = decimal + (digit * powercal(2, power));
        power++;
        binary_num = binary_num / 10;
    }
    return decimal;
}

int decimalToBinary(int num){
    int remainder, place, binary;
    place = 1;
    binary = 0;
    while (num > 0){
        remainder = num % 2;
        binary = binary + remainder * place;
        num = num / 2;
        place = place * 10;
    }

    return binary;
}

int main (){
    int num1, num2, deci1, deci2, result;

    cout << "Enter the BINARY number 1 : ";
    cin >> num1;
    cout << "Enter the BINARY number 2 : ";
    cin >> num2;
```

```
   deci1 = binaryToDecimal(num1); // converting binary to
decimal
   deci2 = binaryToDecimal(num2); // converting binary to
decimal

   result = deci1 + deci2;

   result = decimalToBinary(result);

   cout << "Addition of " << num1 << " and " << num2 << " is :
" << result;
   return 0;
}
```

Output:

```
(base) abhinyaay@Abhinays-MacBook-Air cpp  % ./binaryadd.o
Enter the BINARY number 1 : 1001
Enter the BINARY number 2 : 1100
Addition of 1001 and 1100 is : 10101%
(base) abhinyaay@Abhinays-MacBook-Air cpp  %
```

Add two characters:

Program :

```
#include<iostream>
using namespace std;

int main (){
   char char1 , char2 ;
   int result;
   cout<<"Enter CHAR 1 : ";
   cin>>char1;
   cout<<"Enter CHAR 2 : ";
   cin>>char2;
   result  = char1 + char2;
```

```
    cout <<"Additon of " << char1 << " and " << char2 << " is :
" << result;



    return 0;
}
```

Output:

```
(base) abhinyaay@Abhinays-MacBook-Air cpp  % ./addchar.o
Enter CHAR 1 : c
Enter CHAR 2 : d
Additon of c and d is : 199%
(base) abhinyaay@Abhinays-MacBook-Air cpp  %
```

3. Calculate compound interest:

Program:

```
#include<iostream>
using namespace std;



// power function
double powercal(double base, int expo){
    double num = 1;
    for(int i = 0; i < expo; i++){
        num = num * base;
    }
    return num;
}

int main (){
    int principal;
    int time;
    float rate;
    double amount, compound_interest;
```

```
    cout << "Enter the PRINCIPAL : ";
    cin >> principal;

    cout << "Enter the RATE : ";
    cin >> rate;

    cout << "Enter the TIME (in years): ";
    cin >> time;

    amount = principal * powercal((1 + rate/100), time);

    compound_interest = amount - principal;

    cout << "COMPOUND INTEREST is : " << compound_interest;

    return 0;
}
```

Output:

```
(base) abhinyaay@Abhinays-MacBook-Air cpp  % ./compundIntrest.o
Enter the PRINCIPAL : 10000
Enter the RATE : 2.6
Enter the TIME (in years): 2
COMPOUND INTEREST is : 526.76%
(base) abhinyaay@Abhinays-MacBook-Air cpp  %
```

4. Calculate power of a number:

Program:

```
#include <iostream>
using namespace std;

int powercal(int base, int expo){
    int num = 1;
    for( int i=0; i<expo ; i++ ){
```

```
        num = num * base;
    }
    return num;
}

int main(){
    int base , expo, result;
    cout << "Enter BASE : ";
    cin >> base;
    cout << "Enter EXPONENT : ";
    cin >> expo;
    result = powercal(base, expo);

    cout << " RESULT is " << result;
    return 0;
}
```

Output:

```
(base) abhinyaay@Abhinays-MacBook-Air cpp  % ./powerofNumber.o
Enter BASE : 10
Enter EXPONENT : 3
 RESULT is 1000%
(base) abhinyaay@Abhinays-MacBook-Air cpp  %
```

5. Swap two numbers:

Program:

```
#include <iostream>
using namespace std;

// swapping function
void swapNumbers(int &a, int &b) {
    int temp;
    temp = a;
    a = b;
    b = temp;
```

```
}

int main() {
    int x, y;

    cout << "Enter two numbers: ";
    cin >> x >> y;

    swapNumbers(x, y);

    cout << "After swapping: " << x << " " << y;

    return 0;
}
```

Output :

```
(base) abhinyaay@Abhinays-MacBook-Air cpp  % ./swapNumbers.o
Enter two numbers: 24
45
After swapping: 45 24%
(base) abhinyaay@Abhinays-MacBook-Air cpp  %
```

6. Calculate area of rectangle:

Program :

```
#include<iostream>
using namespace std;

// function to calculate area
int areaRectangle(int length, int breadth){
    return length * breadth;
}

int main(){
    int length, breadth, area;
```

```
    cout << "Enter length: ";
    cin >> length;

    cout << "Enter breadth: ";
    cin >> breadth;

    area = areaRectangle(length, breadth);

    cout << "Area of rectangle = " << area;

    return 0;
}
```

Output:

```
(base) abhinyaay@Abhinays-MacBook-Air cpp  % ./areaRectangle.o
Enter length: 56
Enter breadth: 77
Area of rectangle = 4312%
(base) abhinyaay@Abhinays-MacBook-Air cpp  %
```

7. Get 4 digit number from user, write a function to calculate sum of digits of the 4-digit number:
Using loop -

Program :

```
#include<iostream>
using namespace std;

// function to calculate sum of digits
int sumOfDigits(int num){
    int digit, sum = 0;

    while(num > 0){
        digit = num % 10;    // get last digit
        sum = sum + digit;   // add digit to sum
        num = num / 10;      // remove last digit
```

```
    }

    return sum;
}

int main(){
    int number, result;

    cout << "Enter a 4-digit number: ";
    cin >> number;

    result = sumOfDigits(number);

    cout << "Sum of digits = " << result;

    return 0;
}
```

Output:

```
(base) abhinyaay@Abhinays-MacBook-Air cpp  % ./fourdigitsumusingloop.o
Enter a 4-digit number: 1234
Sum of digits = 10%
(base) abhinyaay@Abhinays-MacBook-Air cpp  %
```

Get 4 digit number from user, write a function to calculate sum of digits of the 4-digit number:
Using recursion -

Program :

```
#include<iostream>
using namespace std;

// function for sum of digit
int sumOfDigits(int num){
    if(num == 0)
        return 0;
```

```
    return (num % 10) + sumOfDigits(num / 10);
}

int main(){
    int number, result;

    cout << "Enter a 4-digit number: ";
    cin >> number;

    result = sumOfDigits(number);

    cout << "Sum of digits = " << result;

    return 0;
}
```

Output:

```
(base) abhinyaay@Abhinays-MacBook-Air cpp  % ./fourdigitsumusingrecursion.o
Enter a 4-digit number: 5674
Sum of digits = 22%
(base) abhinyaay@Abhinays-MacBook-Air cpp  %
```

8. Write a recursive function to obtain n numbers of a Fibonacci series
0 1 1 2 3 5 8 13 21 34 55 89...

Program :

```
#include<iostream>
using namespace std;

// function for fibonacci
int fibonacci(int n){
    if(n == 0)
        return 0;
    else if(n == 1)
        return 1;
```

```
    else
        return fibonacci(n-1) + fibonacci(n-2);
}

int main(){
    int n;

    cout << "Enter number of elements: ";
    cin >> n;

    for(int i = 0; i < n; i++){
        cout << fibonacci(i) << " ";
    }

    return 0;
}
```

Output :

```
(base) abhinyaay@Abhinays-MacBook-Air cpp  % ./fibonacci.o
Enter number of elements: 12
0 1 1 2 3 5 8 13 21 34 55 89 %
(base) abhinyaay@Abhinays-MacBook-Air cpp  %
```

9. Write a function to find the binary equivalent of this number using Recursion.

Program :

```
#include <iostream>
using namespace std;

// recursive function
void decimalToBinary(int n) {
    if(n == 0)
        return;

    decimalToBinary(n / 2);    // recursive call
```

```
    cout << n % 2;              // print remainder
}

int main() {
    int num;

    cout << "Enter a decimal number: ";
    cin >> num;

    cout << "Binary equivalent: ";
    decimalToBinary(num);

    return 0;
}
```

Output :

```
(base) abhinyaay@Abhinays-MacBook-Air cpp  % ./binaryequivalent.o
Enter a decimal number: 11
Binary equivalent: 1011%
(base) abhinyaay@Abhinays-MacBook-Air cpp  %
```